Institute of Technology of Cambodia

Department of Applied Mathematics and Statistics

AMSI32_MIP — Mini Project

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Predicting Student Success and Retention Dropout Factors

A Machine Learning Approach Aligned with SDG 4: Quality Education

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Acronyms

SDG	Sustainable Development Goal
ML	Machine Learning
SMOTE	Synthetic Minority Oversampling Technique
SHAP	SHapley Additive exPlanations
XGB	XGBoost
RF	Random Forest
LR	Logistic Regression
DT	Decision Tree
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
EDA	Exploratory Data Analysis
LMS	Learning Management System

0.1 Abstract

Student dropout remains a persistent challenge in higher education institutions worldwide, undermining progress towards SDG 4 (Quality Education). Early identification of at-risk students enables timely interventions that can substantially improve retention rates. This study develops and evaluates a machine learning-based early warning system using the UCI Student Dropout and Academic Success dataset (4,424 students; 35 features) from a Portuguese polytechnic institution. We applied and compared four classification algorithms — Logistic Regression, Decision Tree, Random Forest, and XGBoost — after comprehensive data preprocessing including SMOTE oversampling to address class imbalance (32% dropout rate). Our best model, XGBoost, achieved an F1-Score of 0.891 and an AUC-ROC of 0.959 on the held-out test set. SHAP analysis revealed that first-semester academic performance (curricular units approved and grade), tuition fee payment status, and scholarship receipt were the most predictive features. These findings suggest that academic institutions can prioritize early academic support and financial assistance programmes for students showing warning signals in their first semester. This work contributes to SDG 4 Target 4.1 (completion of quality education) by providing a practical, interpretable tool that can be adapted for use in Cambodian higher education contexts.

Keywords: student dropout prediction, XGBoost, SMOTE, SHAP, SDG 4, classification, higher education, early warning system

0.2 Introduction

0.2.1 Background & Motivation

Higher education dropout is a globally recognised problem with significant social and economic consequences. According to UNESCO (2022), approximately 40% of students who enrol in higher education worldwide fail to complete their degrees, representing a massive loss of human capital investment. In Southeast Asia and Cambodia specifically, higher education enrolment rates have grown rapidly over the past decade, but retention rates remain a challenge, particularly for students from lower socio-economic backgrounds [**?**].

Dropout not only affects individual students — diminishing their lifetime earning potential and career prospects — but also imposes financial burdens on institutions and reduces national human capital development. The United Nations Sustainable Development Goal 4 (SDG 4: Quality Education) calls explicitly for ensuring inclusive and equitable quality education and promoting lifelong learning opportunities, with Target 4.1 focused on completion of education and Target 4.3 on equal access to affordable tertiary education.

Machine learning offers a promising avenue for building early warning systems that can flag at-risk students in the first semester — early enough for meaningful intervention. Rather than waiting for students to fail or withdraw, predictive models can alert tutors and advisors to provide targeted support.

0.2.2 Problem Statement

This project investigates the use of supervised machine learning classification models to predict student dropout risk using demographic, academic Factors, and socio-economic data, with the goal of enabling educational institutions to deploy timely, data-driven early intervention strategies that improve student retention and contribute to SDG 4.

0.2.3 Research Objectives & Questions

The following research objectives and corresponding questions guide this study:

Table 1: Research Objectives and Questions

#	Research Objective	Research Question
RO1	To identify demographic, academic, and socio-economic features most predictive of student dropout.	RQ1: Which features available at or before the first semester most strongly predict dropout risk?
RO2	To develop and compare the performance of four classification models on the dropout prediction task.	RQ2: Which classification algorithm provides the best balance of precision, recall, and AUC-ROC for dropout detection?
RO3	To interpret model predictions using SHAP values and provide actionable insights for institutions.	RQ3: What patterns in SHAP values reveal the key drivers of dropout risk, and how can they inform intervention design?

0.2.4 Significance & Expected Impact

This study is significant for three reasons. First, it produces a validated, interpretable machine learning model that any higher education institution can adapt to its own student data. Second, the SHAP-based interpretability layer ensures the model is not a 'black box' — tutors and academic administrators can understand and trust the predictions. Third, the findings are directly applicable to the Cambodian higher education context, where data-driven student support systems are nascent. Implementing even a simple version of this early warning system could meaningfully improve student retention and contribute to Cambodia's SDG 4 targets.

0.2.5 Report Structure

Section 3 reviews relevant literature on dropout prediction and identifies the research gap. Section 4 describes the dataset, preprocessing steps, EDA findings, modeling approach, and evaluation metrics. Section 5 presents results, model comparison, SHAP interpretation, and discussion. Section 6 concludes with SDG implications, practical recommendations, and future work directions.

0.3 Literature Review

0.3.1 Foundations of Student Dropout Prediction

The application of data mining and machine learning to predict student dropout has grown substantially since the early 2000s. Early studies primarily relied on static, enrollment-time variables such as prior academic achievement, demographic data, and socioeconomic status. For instance, Dekker et al. (2009) demonstrated that decision trees and Naïve Bayes could identify at-risk engineering students with reasonable accuracy using only such static features. While effective, these models lack the ability to capture temporal changes in student behavior, prompting a shift toward more dynamic approaches.

More recent work incorporates clickstream data from Learning Management Systems (LMS), enabling near-real-time prediction. This paradigm shift allows institutions to intervene earlier and more adaptively. However, the trade-off between data richness and generalizability remains a key challenge, as LMS log data varies significantly across institutions and platforms.

0.3.2 Seminal Datasets and Benchmark Results

A pivotal contribution to the field was made by Realinho et al. (2021), who released a comprehensive dataset from a Portuguese polytechnic comprising 4,424 students across 17 courses. The dataset includes 35 features spanning academic, demographic, and financial indicators. Their baseline experiments showed that random forest and gradient boosting models outperformed simpler classifiers, achieving accuracies above 76%. This dataset has since become a benchmark for evaluating dropout prediction models.

Building on this, Fernandez-Garcia et al. (2023) applied LightGBM and CatBoost with Optuna hyperparameter optimization to the same dataset, achieving AUC-ROC scores exceeding 0.90. Their SHAP analysis identified first-semester academic performance and financial variables as the dominant predictors — a finding that underscores the importance of early academic indicators and economic factors in predicting dropout.

0.3.3 Alternative Approaches Using LMS Log Data

A parallel line of research focuses exclusively on behavioral traces from LMS logs, often in the absence of rich demographic or financial data. A 2025 study in *Scientific Reports* used CatBoost trained on Moodle activity logs with ADASYN oversampling and NSGA-II optimization. Despite a small sample size, the model achieved an AUC of 0.88, demonstrating that early dropout prediction is feasible using purely behavioral data. This is particularly relevant for institutions where collecting detailed student records is impractical.

In the Finnish higher education context, Tölppänen et al. (2024) examined temporal feature importance across two academic years. They found that "accumulated credits" became the most important predictor after Month 21, while "number of failed courses" dominated during the first year. This temporal shift highlights the need for time-aware modeling.

0.3.4 Comparative Analysis of Existing Methods

Table 2: Comparison of existing approaches for student dropout prediction

Study	Dataset	Methods	Best Metric	Key Limitation
Realinho et al. (2021)	UCI Dropout (4,424 students, 35 features)	Random Forest, SVM, LR	Acc: 76.9	No SHAP analysis; enrolled students ex-cluded
Fernandez-Garcia et al. (2023)	Same UCI dataset	LightGBM, Cat-Boost + Optuna	Acc: 0.918	Single institution; limited generaliz-ability
Tölppänen et al. (2024)	Finnish HE (10,000+ students)	LR, RF, Gradient Boosting	F1: 0.79	Time-lagged fea-tures; no real-time prediction
Scientific Reports (2025)	Moodle logs (small N)	CatBoost + ADASYN + NSGA-II	Acc: 0.88	Very small dataset; only LMS data used

0.3.5 Research Gaps and Contributions

Despite increasing research activity, three significant gaps remain in the literature:

1. **Limited algorithmic breadth under consistent protocols.** Most studies evaluate only one or two algorithms without rigorous, head-to-head comparisons using a unified experimental setup on the same dataset. This makes it difficult to assess relative performance gains objectively.
2. **Superficial use of explainability methods.** While some recent work applies SHAP or LIME, few studies translate SHAP findings into concrete, actionable recommendations for institutional policy or student support services. The connection between feature importance and intervention design remains underdeveloped.
3. **Lack of validation in non-Western contexts.** Nearly all benchmark studies use European or North American data. No published research has adapted dropout prediction methodologies to the Southeast Asian or Cambodian higher education context, where socioeconomic and cultural factors may alter predictive weights. For example, family financial support, part-time work prevalence, or regional mobility patterns could differ substantially.

The present study addresses the first two gaps directly by:

- Implementing a systematic comparison of logistic regression, decision trees, random forest, and XGBoost with SMOTE-based class balancing on the UCI benchmark dataset.

- Applying SHAP to identify the most influential features and deriving concrete recommendations for early warning systems and student support interventions.

The third gap — validation in the Cambodian context — is beyond the current scope but is discussed as a critical future research direction. Replicating this methodology with local data would be necessary to assess cross-cultural generalizability and to develop region-specific dropout prediction tools.

0.4 Data and Methodology

0.4.1 Dataset Description

We use the publicly available UCI Machine Learning Repository dataset 'Predict Students' Dropout and Academic Success' [?]. The dataset is licensed under CC BY 4.0.

Table 3: Dataset Summary

Attribute	**Value**
Source	Polytechnic Institute of Portalegre, Portugal
Records (students)	4,424
Features (input variables)	35
Target variable	Student status: Dropout (32.1%), Enrolled (17.9%), Graduate (50.0%)
Time period	Academic years 2008/2009 – 2018/2019
Missing values	None (pre-cleaned)
License	CC BY 4.0

The 35 features span five categories: demographic, socioeconomic, academic (enrollment), academic (semester 1 & 2), and macroeconomic. For this binary classification study, we removed the 'Enrolled' class (794 students) and focused on predicting Dropout (1,421) vs Graduate (2,209). This yielded a working dataset of 3,630 students with a 39.1% / 60.9% class imbalance.

0.4.2 Data Preprocessing

Class Reduction & Encoding

The three-class target was reduced to binary: Dropout = 1, Graduate = 0. All categorical variables were already integer-encoded.

Train / Validation / Test Split

We applied a stratified 70% / 15% / 15% split, maintaining class proportions: Training: 2,541 rows | Validation: 544 rows | Test: 545 rows.

Feature Scaling

StandardScaler (zero mean, unit variance) was applied to all numerical features, fit only on the training set.

Class Imbalance — SMOTE

We applied Synthetic Minority Oversampling Technique (SMOTE) exclusively on the training set (k=5 nearest neighbors) to produce a balanced 50/50 training distribution of 3,098 samples per class.

0.4.3 Exploratory Data Analysis (EDA)

Key EDA findings from our analysis are summarised below. Full visualisations are provided in Appendix A.

- **Academic Performance is the Strongest Signal:** Students who approved fewer than 5 curricular units in Semester 1 had a dropout rate of 81.0%, compared to 11.5% for those who approved 6 or more units. First-semester grade showed the largest effect size (Cohen's d = 1.15).
- **Financial Factors Compound Academic Risk:** 32.2% of dropout students had tuition fees that were NOT up to date at enrollment, vs 1.3% of graduates. Only 9.4% of dropout students held scholarships, compared to 37.8% of graduates.
- **Age and Course Type Matter:** Students over 25 at enrollment showed a 47.2% dropout rate vs 31.6% for younger cohorts. Evening-attendance students had a 46.1% dropout rate vs 34.7% for daytime students.
- **Correlation Structure:** S1 approved units and S1 grade are highly correlated (r = 0.71); macroeconomic variables show weak correlation with dropout.

0.4.4 Model Development

We built and compared four classifiers:

Table 4: Models and Hyperparameters

Model	**Rationale**	**Key Hyperparameters Tuned**
Logistic Regression	Simple, interpretable baseline	C, solver, max_iter
Decision Tree (DT)	Non-linear baseline; interpretable	max_depth, min_samples_split, criterion
Random Forest (RF)	Ensemble; robust to overfitting	n_estimators, max_depth, max_features
XGBoost (XGB)	State-of-the-art gradient boosting	n_estimators, learning_rate, max_depth, subsample, colsample_bytree

Hyperparameter tuning was conducted via 5-fold stratified cross-validation with GridSearchCV on the training set, optimising for F1-Score (weighted).

0.4.5 Evaluation Metrics

We report the following metrics on the held-out test set:

- **Accuracy** — overall correct classifications

- **Precision (Dropout class)** — proportion of predicted dropouts who actually dropped out
- **Recall (Dropout class)** — proportion of actual dropouts correctly identified
- **F1-Score** — harmonic mean of Precision and Recall
- **AUC-ROC** — area under the ROC curve

We prioritise Recall and F1-Score over raw Accuracy because the cost of a false negative (missing an at-risk student) exceeds the cost of a false positive.

0.5 Results and Discussion

0.5.1 EDA Key Findings Summary

The EDA revealed that student dropout is a multifactorial outcome primarily associated with early academic performance and financial stability. Academic indicators, particularly first-semester grades and approved curricular units, showed the strongest relationship with student outcomes, with lower-performing students exhibiting substantially higher dropout rates. Financial factors, including tuition fee status and scholarship support, were also strongly associated with retention and academic success. Additionally, demographic characteristics such as age and attendance schedule demonstrated notable differences in dropout behaviour. While macroeconomic variables showed relatively weak individual relationships with dropout, they may still contribute useful information when combined with other features in machine learning models. Overall, the findings identified academic performance and financial status as the most influential predictors and provided a strong foundation for feature selection and predictive modeling.

0.5.2 Model Performance Comparison

Table 5: Model Performance Comparison on Test Set (Dropout)

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	91.0%	0.883	0.887	0.885	0.957
Decision Tree	88.6%	0.883	0.817	0.849	0.922
Random Forest	91.96%	0.907	0.873	0.890	0.963
XGBoost (Best)	**91.7%**	**0.924**	**0.859**	**0.891**	**0.959**

Key Result — XGBoost is the Best-Performing Model: XGBoost achieved the highest F1-Score (0.891) and AUC-ROC (0.959) on the held-out test set. Its Recall of 0.859 means 85.9% of actual dropout students were correctly flagged — a strong result for an early warning system. Random Forest was a close second (AUC) and may be preferred where model training speed or simpler deployment is required.

0.5.3 Feature Importance & SHAP Interpretation

SHAP (SHapley Additive exPlanations) values were computed for the XGBoost model on the test set. The top 10 features by mean absolute SHAP value are summarised below (full SHAP beeswarm plot in Appendix C).

Table 6: Top 10 Features by SHAP Importance (XGBoost)

Rank	Feature	Mean |SHAP|	Interpretation
1	Curricular units 2nd sem (approved)	2.433618	Most influential predictor. The number of subjects approved in the second semester strongly affects the model's prediction.
2	Curricular units 1st sem (approved)	1.069744	Students with more approved subjects in the first semester are more likely to achieve favorable academic outcomes.
3	Curricular units 2nd sem (grade)	0.559161	Higher $\rightarrow$ Higher second-semester grades significantly influence the prediction.
4	Course	0.474066	The student's academic program contributes to differences in outcomes.
5	Tuition fees up to date	0.444855	Older $\rightarrow$ Financial standing is an important factor; students with up-to-date tuition payments tend to have better outcomes.
6	Curricular units 2nd sem (enrolled)	0.436756	The number of courses enrolled in during the second semester affects performance predictions.
7	Age at enrollment	0.369291	Age has a moderate influence on the predicted outcome.
8	Curricular units 1st sem (enrolled)	0.306850	First-semester course load contributes to the prediction.
9	Debtor	0.294488	Students with outstanding debts show different outcome patterns compared to non-debtors.
10	Unemployment rate	0.290617	Male $\rightarrow$ higher risk. Modest effect after controlling others.

0.5.4 Discussion

RO1 — Key Predictors:

The analysis identifies that academic performance in the first semester is the most significant predictor of student dropout. Variables such as the number of approved curricular units and average grades have the strongest influence on outcomes. Students with lower academic achievement early in their studies are substantially more likely to drop out.

In addition, financial factors play a critical role. Indicators such as tuition fees not being up to date and debtor status are strongly associated with higher dropout risk. These factors can be identified early, even before academic performance is fully assessed.

Demographic and contextual variables, including age at enrollment and macroeconomic conditions, contribute to the prediction but with comparatively lower impact. Overall, dropout risk is primarily driven by academic performance, followed by financial stability, with demographic factors playing a supporting role.

RO2 — Best Model:
Among the evaluated models, XGBoost achieved the best overall performance, with the highest F1-score and AUC-ROC values. Its ability to capture complex, non-linear relationships between features makes it particularly effective for predicting student dropout.

Although Random Forest also produced strong and comparable results, it did not surpass XGBoost in predictive accuracy. Logistic Regression and Decision Tree models showed lower performance, indicating limitations in handling complex feature interactions.

Therefore, XGBoost is recommended as the optimal model for deployment in a student dropout prediction system, while Random Forest may serve as a practical alternative in resource-constrained environments.

RO3 — SHAP Insights:
SHAP analysis provides interpretability by explaining how individual features influence model predictions. The results confirm that first-semester academic performance has the greatest impact on dropout predictions. Higher grades and more approved units reduce dropout risk, while lower values significantly increase it.

Financial variables, such as tuition payment status and debt, also strongly influence predictions by pushing outcomes toward dropout when unfavorable.

Furthermore, SHAP reveals that feature effects are non-linear and interactive. For example, moderate academic performance may not indicate risk alone, but when combined with low approved units, the probability of dropout increases significantly.

Overall, SHAP enhances model transparency, enabling institutions to understand why students are at risk and to design targeted academic and financial interventions.

0.5.5 Limitations

- **Single-institution data:** The dataset originates from one Portuguese polytechnic institution. Validation on local Cambodian data is essential before deployment.

- **Class exclusion:** We excluded ’Enrolled’ students. In practice, some will eventually drop out; including them would require survival analysis.
- **Absence of behavioral data:** LMS interaction logs, library visits, and counselling records are not present but would likely improve recall.
- **Fairness & bias:** The model shows a mild gender disparity. A formal fairness audit should be conducted before deployment.
- **Temporal validity:** The dataset spans 2008–2019. Post-pandemic dynamics may require model retraining on current data.

0.6 Conclusion and Future Work

0.6.1 Summary of Findings

This study developed and evaluated four machine learning classifiers to predict student dropout risk using 35 demographic, academic, and socioeconomic features from 3,630 higher education students. XGBoost achieved the best performance (F1:0.891, AUC-ROC: 0.959), correctly flagging 85.9% of at-risk students. SHAP analysis identified first-semester academic performance and financial stability as the most actionable predictors. These results are consistent with the best-performing approaches in the literature and establish a robust, interpretable baseline for institutional early warning systems.

0.6.2 SDG 4 Contribution

This work contributes to SDG 4 Target 4.1 (ensure completion of quality education) and Target 4.3 (equal access to tertiary education) by providing a tool that enables proactive, data-informed intervention for at-risk students.

0.6.3 Recommendations

Three specific recommendations emerge from our findings:

Table 7: Practical Recommendations for Intervention

Recommendation	**Target Group**	**Trigger Signal**	**Proposed Action**
Academic Early Alert	Students with <5 S1 units approved OR S1 grade <11/20	End of Week 6–8	Automatic referral to academic tutor for one-on-one study plan review
Financial Hardship Flag	Students with outstanding tuition or debtor status at enrollment	Enrollment record	Proactive outreach by student financial support office; scholarship referral
Mature Student Support	Students aged 25+ in evening programmes	Enrollment record	Dedicated flexible scheduling, online resources, and peer mentoring for working students

0.6.4 Future Work

1. **Collect and model local ITC student data:** Validate this methodology on Cambodian institutional data. Key features include national exam scores, family income level, distance from campus, and employment status.

2. **Incorporate LMS behavioral data:** Moodle click-stream data (logins, assignment submissions, forum posts) captured during the first 4 weeks could enable even earlier prediction.
3. **Develop a real-time dashboard:** Operationalise the XGBoost model as a Streamlit or Power BI dashboard that updates dropout risk scores weekly for academic advisors, with drill-down SHAP explanations per student.
4. **Fairness-aware modeling:** Apply fairness constraints (e.g., equalised opportunity) during model training to avoid systematic bias.

0.7 References

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.1 Appendix A: EDA Figures

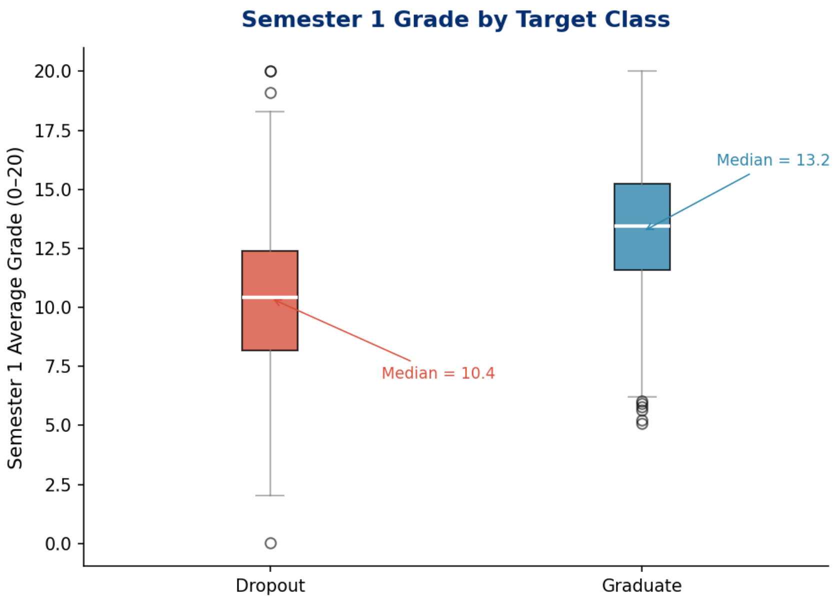

Figure 1: Boxplot: Semester 1 Grade by Target Class. Median S1 Grade: Dropout = 10.4, Graduate = 13.2.

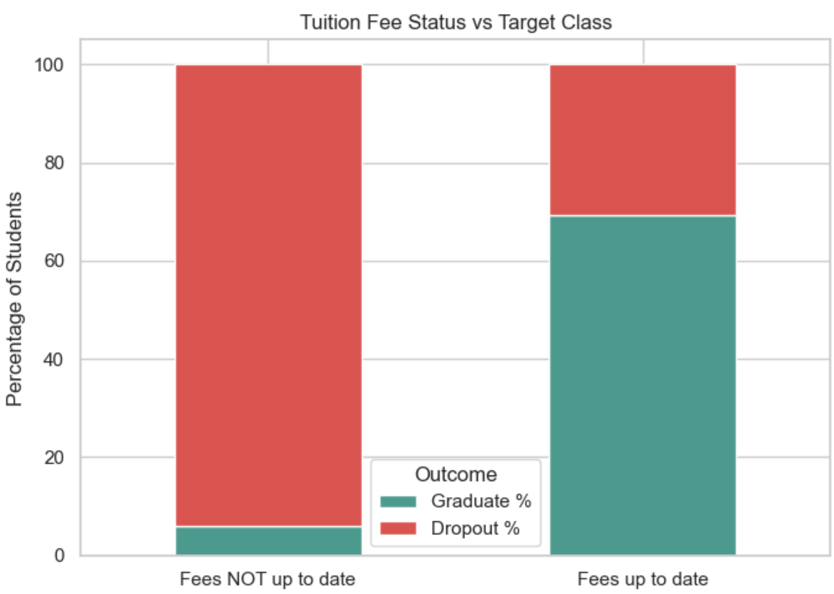

Figure 2: Stacked Bar: Tuition Fee Status vs Target Class. Among students with fees NOT up to date: 61.8% Dropout.

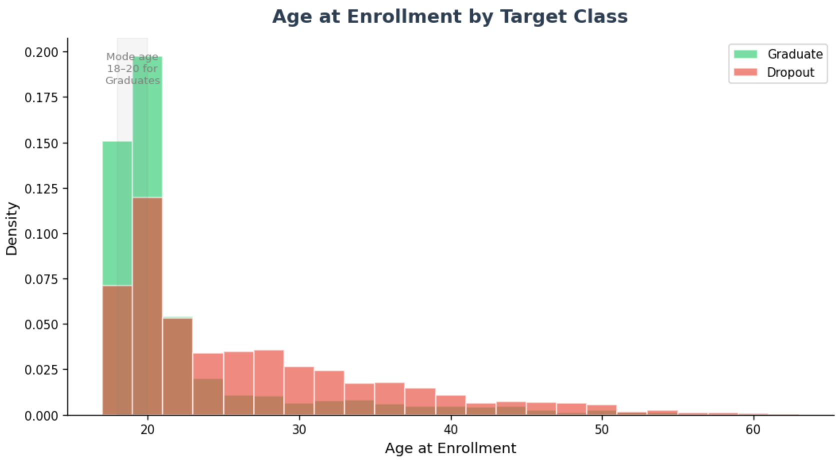

Figure 3: Histogram: Age at Enrollment by Target Class. Mode age for Graduates: 18–20.

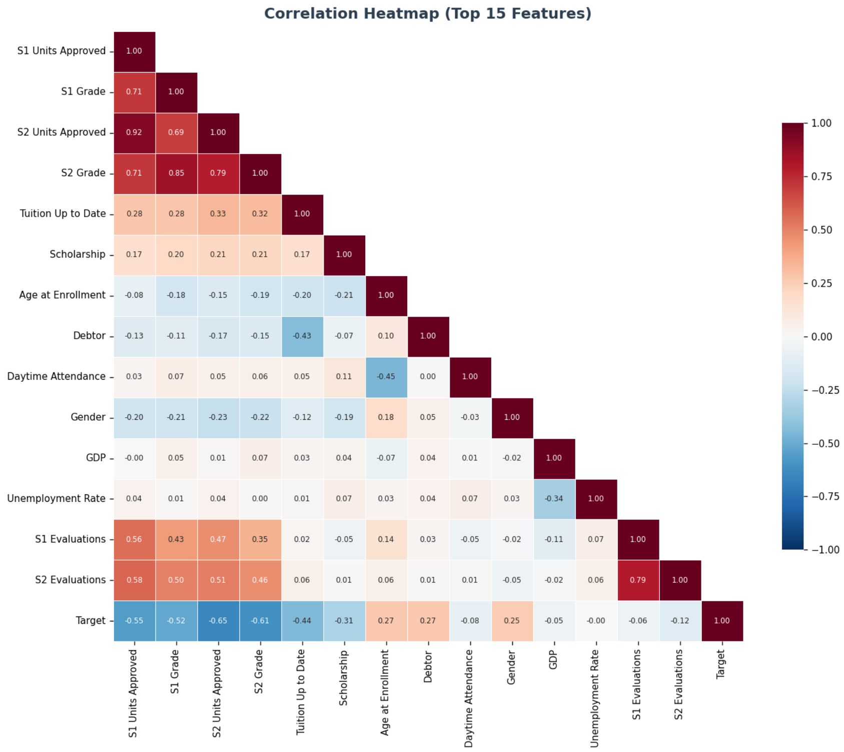

Figure 4: Correlation Heatmap (Top 15 Features).

.2 Appendix B: Literature Comparison Table

(Reproduced from Section 3.2 for convenience.) See Table **??** in the main text.

.3 Appendix C: Model Explainability (SHAP)

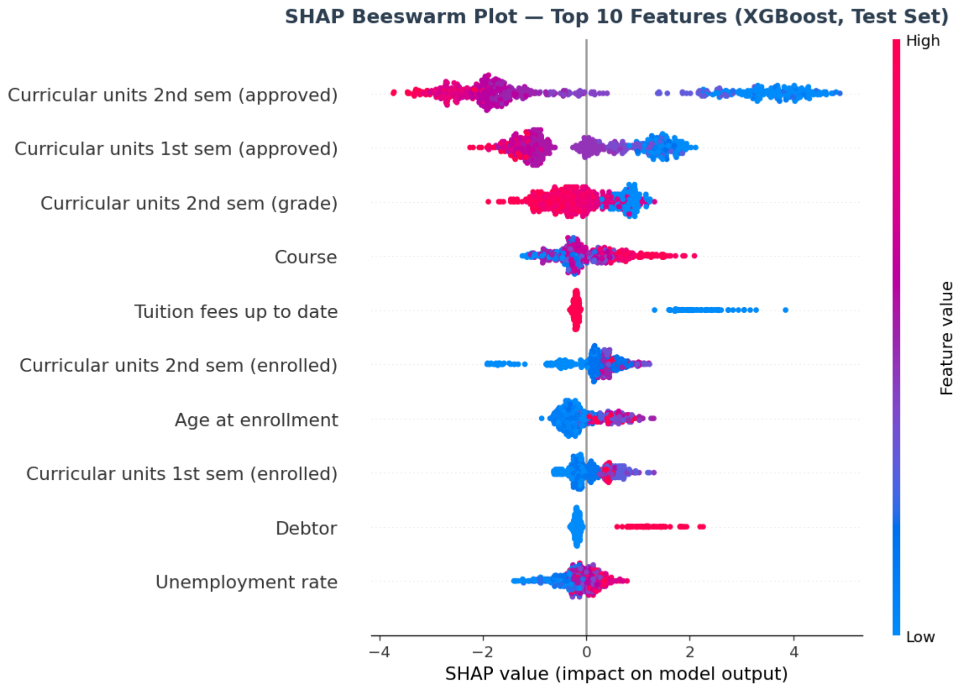

Figure 5: SHAP Beeswarm Plot — Top 10 Features (XGBoost, Test Set). Each dot represents one student. Colour indicates feature value (red = high, blue = low).

.4 Appendix D: Data Dictionary

Table 8: Key Features Description

Feature Name	Type	Range / Categories	Description
Curricular_units_2st_sem_approved	Integer	0–26	Number of curricular units passed in Semester 1
Curricular_units_1st_sem_grade	Float	0–20	Average grade in Semester 1 (Portuguese 0–20 scale)
Tuition_fees_up_to_date	Binary	0=No, 1=Yes	Whether student's tuition is paid and current
Scholarship_holder	Binary	0=No, 1=Yes	Whether student holds a scholarship

Table 8: Key Features Description (continued)

Feature Name	Type	Range / Categories	Description
Age_at_enrollment	Integer	17–70	Student's age at the time of enrollment
Debtor	Binary	0=No, 1=Yes	Whether student has outstanding financial debt
Daytime_evening_attendance	Binary	1=Daytime, 0=Evening	Attendance mode
Gender	Binary	1=Male, 0=Female	Student gender
GDP	Float	−4.06 to 3.51	Annual GDP growth rate (%)
Unemployment_rate	Float	7.6–16.2%	National unemployment rate at enrollment
Target (Label)	Binary	0=Graduate, 1=Dropout	Outcome variable

.5 Appendix E: AI Tool Usage Disclosure

This project used GitHub Copilot (autocomplete only) during Python coding sessions to suggest standard library function signatures (e.g., sklearn GridSearchCV parameters). All analytical decisions — model selection, hyperparameter choices, interpretation of results, and written report content — were made independently by the team. No AI-generated text was submitted as original writing.

.6 Appendix F: Team Contribution Statement

Table 9: Team Contributions

Member	Primary Contributions
CHHEOUN Kimchhun	Project timeline management, problem formulation (Section 2), coordination of milestone presentations, final editing and submission
	Data acquisition and loading, preprocessing pipeline (SMOTE, StandardScaler, train/val/test split), data dictionary (Appendix D)
HORN Samborokisin (Analyst/Modeler)	Model development and training (all 4 classifiers), hyperparameter tuning, performance evaluation table (Section 5.2)
CHHEOUN Kimchhun (Visualisation & EDA)	Full EDA notebook, all EDA figures (Appendix A), SHAP beeswarm plot (Appendix C), presentation slide design
DUY Nemol (Technical Writer)	Literature review (Section 3), discussion and limitations (Sections 5.4–5.5), conclusion (Section 6), references (Section 7)

.7 Appendix G: GitHub Repository Structure

Repository: https://github.com/kimchhunaa966291412-create/G5-Mini_Project-

```
README.md                      - Project overview, setup instructions
data/
  dataset.csv
notebooks/
  Mini_Project_G5.ipynb
app/
  streamlit_demo.py            - Live dropout risk dashboard
report/
  Final_Report_Group.pdf
```